Time Value of Money and Net Present Value

CHEME 5660
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LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Common units.** Explain why cash flows occurring at different times cannot be added directly and how discounting makes them comparable.
- **Compounding conventions.** Derive discrete and continuous discount factors from a stated compounding rule.
- **Asset valuation.** Represent a project, product, or security as an abstract asset and compute its net present value.

1 Why Does Time Have a Price?

The time value of money begins with a simple asymmetry: one dollar in hand today and one dollar promised next year carry the same currency label, but they do not offer the same set of choices. Today's dollar can be consumed, held as a reserve, or invested. The future dollar cannot be used until it arrives, and its purchasing power and delivery are uncertain.

This idea predates modern financial markets. Irving Fisher organized it around two forces: impatience to spend income and the opportunity to invest it. Modern valuation adds inflation and risk to the same basic story. A discount rate is therefore not merely a computational input; it records the return forgone by waiting and, when appropriate, compensation for uncertainty.

A useful currency model treats money at different dates as different currencies: $\mathrm{USD}_0, \mathrm{USD}_1, \dots$. Just as euros and U.S. dollars cannot be added before conversion, dated dollars cannot be added until they share a valuation date. The discount rate specifies the conversion across time, and the discount factor performs it. An abstract asset then provides a flexible model of the economic value of a process, good, or idea as dated signed cash flows (Fig. 1).

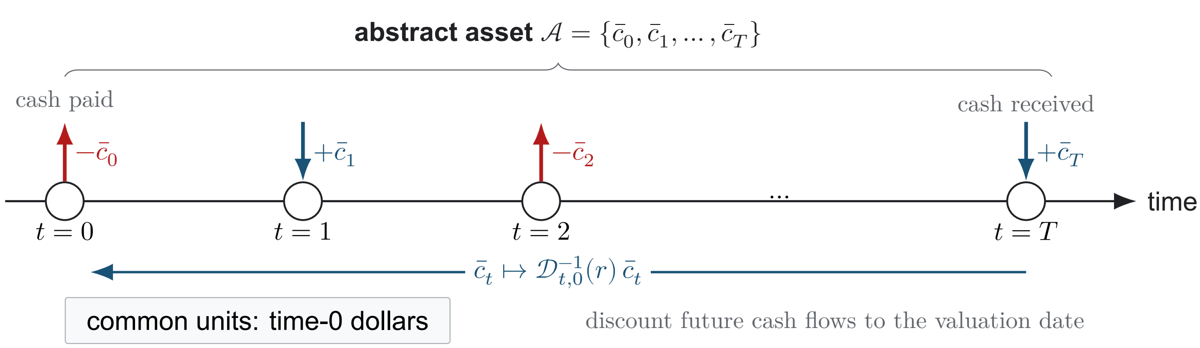

Figure 1: The abstract asset $\mathcal{A}$ comprises a signed net cash flow $\bar{c}_t$ at each time $t \in \{0, 1, \dots, T\}$, where the terminal time T is the asset life. Following the process-flow convention, an arrow entering a time-point unit is money received and an arrow leaving it is money paid. Discounting transports each future cash flow to the common valuation date $t = 0$.

2 Discounting as a Change of Units

To construct the required conversion rule, begin with the simplest transport problem. Suppose the initial principal P dollars earns the annual discount rate r for one year. The resulting year-end future value F is given by Eq. 1:

$$F = (1+r)P. \tag{1}$$

With annual compounding, the quoted rate r produces the forward conversion factor $1+r$ and the backward factor $(1+r)^{-1}$. These two directions define a reversible conversion between dated currencies (Defn. 2.1) and let us express every cash flow in Fig. 1 at a common date.

Definition 2.1. Discount factor

Let time 0 be the valuation date, let the horizon $t \geq 0$ identify a future date, and let USD_0 and USD_t denote dollars measured at those respective dates. For a quoted discount rate r and a stated compounding convention, the discount factor $\mathcal{D}_{t,0}(r)$ is the multiplicative conversion factor from USD_0 to USD_t. If the present amount P_0 is measured in USD_0 and the equivalent future amount F_t is measured in USD_t, then $F_t = \mathcal{D}_{t,0}(r)P_0$ and $P_0 = \mathcal{D}_{t,0}^{-1}(r)F_t$, where $\mathcal{D}_{t,0}^{-1}(r) := 1/\mathcal{D}_{t,0}(r)$ is the reciprocal conversion factor. A valid discount factor satisfies $\mathcal{D}_{0,0} = 1$.
In this course, $\mathcal{D}_{t,0}$ denotes the forward accumulation factor. Many finance texts instead reserve the phrase *discount factor* for its present-value inverse, $\mathcal{D}_{t,0}^{-1}$; the formulas are equivalent once the convention is stated.

The periodic-compounding formula is a model rather than an algebraic identity. It assumes that the nominal annual discount rate r remains constant; the positive integer compounding frequency n divides each year into n equal intervals; interest is credited at the end of every interval at the proportional rate r/n; credited interest remains invested; and the investment horizon T years contains the whole number nT of intervals with no intervening deposits or withdrawals.

Let the account value V_k denote the value immediately after interval k, and let the initial principal be $V_0 = P$. Applying the same one-interval update repeatedly gives the periodic accumulation factor:

$$\begin{aligned}
V_0 &= P, && \text{initial deposit}\\
V_1 &= \left(1+\frac{r}{n}\right)V_0 = P\left(1+\frac{r}{n}\right), && \text{apply the interval multiplier once}\\
V_2 &= \left(1+\frac{r}{n}\right)V_1 = P\left(1+\frac{r}{n}\right)^2, && \text{apply it a second time}\\
&\vdots && \text{repeat the same update}\\
V_{nT} &= P\left(1+\frac{r}{n}\right)^{nT}, && \text{after } nT \text{ intervals}\\
\mathcal{D}_{T,0}(r;n) := \frac{V_{nT}}{V_0} &= \left(1+\frac{r}{n}\right)^{nT}, && \text{normalize by the initial value.}
\end{aligned}$$

The exponent nT counts the compounding intervals and the base $1 + r/n$ is the one-interval multiplier; as the compounding frequency n increases, this factor approaches the continuous-compounding limit (Thm. 2.1).

Theorem 2.1. Continuous-compounding limit

Let the nominal annual discount rate be the constant $r \in \mathbb{R}$, where $\mathbb{R}$ denotes the set of real numbers; let the investment horizon be $T \geq 0$ years; and let the positive integer n denote the number of compounding intervals per year. As the compounding frequency n increases without bound, the discrete accumulation factor converges to its continuous counterpart, where e denotes Euler's number, as stated by Eq. 2:

$$\lim_{n\to\infty}\left(1+\frac{r}{n}\right)^{nT} = e^{rT}. \tag{2}$$

Consequently, the continuously compounded discount factor is $\mathcal{D}_{T,0}(r) = e^{rT}$.

Derivation. Apply the natural logarithm log to the discrete accumulation factor $\mathcal{D}_{T,0}(r;n)$ and use the Taylor expansion $\log(1+x) = x - x^2/2 + O(x^3)$, where $x = r/n$ and $O(x^3)$ denotes a remainder bounded in magnitude by $C|x|^3$ near $x = 0$ for some constant $C > 0$:

$$\begin{aligned}\log \mathcal{D}_{T,0}(r;n) &= nT\log\left(1+\frac{r}{n}\right)\\ &= rT - \frac{r^2T}{2n} + O(n^{-2}).\end{aligned}$$

Here $O(n^{-2})$ denotes terms bounded in magnitude by C'/n^2 for sufficiently large compounding frequency n and some constant $C' > 0$. These correction terms vanish as $n \to \infty$; exponentiating gives the result. The expansion also shows how quickly discrete compounding approaches the continuous convention. □

The companion notebook turns these expressions into computed discount factors. Use it to compare discrete and continuous compounding numerically and to see how quickly the two conventions converge.

COMPANION NOTEBOOK
L1b: Time Value of Money →
Build discrete and continuous discount factors and compare the two compounding conventions numerically.

3 Abstract Assets and Net Present Value

With the conversion rule in hand, we can return to the abstract asset. Let $\bar{c}_t$ be the signed net cash flow in USD_t: receipts are positive and payments are negative. An asset with life $0, 1, \dots, T$ is represented by $\{\bar{c}_t\}_{t=0}^{T}$. NPV follows a simple discipline—*convert, then add*: multiply each $\bar{c}_t$ by $\mathcal{D}_{t,0}^{-1}(r)$ to express it in USD_0, then sum the converted values (Defn. 3.1).

Definition 3.1. Net present value

Let time 0 be the valuation date, let the nonnegative integer T be the terminal date, and let each integer $t \in \{0, 1, \dots, T\}$ identify a cash-flow date. Let the signed net cash flow $\bar{c}_t$ be measured in USD_t, and let the discount factor $\mathcal{D}_{t,0}(r)$ convert USD_0 to USD_t under the discount rate r. Its reciprocal $\mathcal{D}_{t,0}^{-1}(r) := 1/\mathcal{D}_{t,0}(r)$ converts the cash flow back to time-0 dollars. The net present value at time 0 is given by Eq. 3:

$$\text{NPV}_0(r) = \sum_{t=0}^{T}\mathcal{D}_{t,0}^{-1}(r)\,\bar{c}_t. \tag{3}$$

Every term in this sum is measured in time-0 currency units.

The sign of NPV compares the asset with the opportunity represented by the discount rate r:

Positive NPV ($\text{NPV} > 0$). Discounted receipts exceed discounted payments. Under the model assumptions, the asset creates value beyond the required return encoded by the discount rate r and passes the NPV decision criterion.

Zero NPV ($\text{NPV} = 0$). The asset earns exactly the required return. The decision maker is economically indifferent between the asset and the benchmark on value grounds, although strategic or operational considerations may still matter.

Negative NPV ($\text{NPV} < 0$). The asset does not earn the required return and fails the NPV criterion. This does not necessarily imply a nominal cash loss: undiscounted receipts may exceed payments while still being insufficient after accounting for time and the benchmark return.

The worked-example notebook now applies the abstract-asset model. It constructs a cash-flow table, computes NPV, and tests how the result changes when the discount rate and other assumptions move.

COMPANION NOTEBOOK

L1b: Net Present Value Worked Example →

Represent a vehicle purchase as an abstract asset, compute its NPV, and explore which assumptions drive the result.

KEY TAKEAWAYS

In this lecture, we placed dated cash flows in common time-0 units, derived discrete and continuous accumulation factors, and used those factors to define NPV for an abstract asset.

- **Time changes units.** Money at different dates behaves like different currencies: the discount rate specifies the conversion, and the discount factor converts each cash flow to a common valuation date.
- **Structure before identity.** An abstract asset separates the universal cash-flow structure from the particular identity of an investment.
- **NPV supports decisions.** NPV sums discounted net cash flows and compares the asset with the opportunity encoded by the discount rate; the notebooks supply the numerical experiments.

Next, use the companion notebooks to test how compounding conventions, discount rates, and cash-flow assumptions change a valuation.

ADDITIONAL READING

- Irving Fisher, *The Theory of Interest* (1930), Chapters I–III, for the classical account of impatience and investment opportunity.
- Jonathan Berk and Peter DeMarzo, *Corporate Finance*, 5th ed. (2020), Chapters 3–4, for present value and investment decision rules.
- Frank J. Fabozzi and Francesco A. Fabozzi, *Bond Markets, Analysis, and Strategies*, 10th ed. (2021), for discount factors and fixed-income conventions.